

PROJECT SHOWCASE & ARCHITECTURE

SHOULD AI BUY ?

Autonomous Multi-Agent Algorithmic Trading with **Adversarial Deliberation**, Verifiable Evidence Ledgers, and Real-Time Broker Execution.

AI COUNCIL ARCHITECTURE

3 Specialist Agents + 1
Adversarial Red Team

EXECUTION ENGINE

Alpaca Paper API v2 with Hard
Risk Gates

PRODUCTION STATUS

24/7 Cloud Daemon on AWS
EC2 (Ubuntu)

The Black-Box Trap in Retail & AI Trading

Why 95% of retail traders lose capital and single LLMs cannot be trusted with real broker balances.

01.
PROBLEM
SPACE

MARKET
FAILURE
ANALYSIS

01

Emotional Capitulation

95% of retail traders fail due to psychological exhaustion: revenge trading, FOMO entries at local tops, and stubborn refusal to honor stops.

Stat: >80% blow up within 90 days.

02

Single LLM Hallucinations

Asking a single ChatGPT or Claude prompt "Should I buy BTC?" generates polite probabilistic text, not quantitative risk modeling or order book awareness.

Flaw: Zero liquidity checks & sycophantic bias.

03

Zero Thesis Explainability

Traditional black-box trading bots flash "BUY" signals without explaining WHY, where invalidation stops lie, or what catastrophic catalysts would void the trade.

Need: Verifiable auditable debate records.

SHOULD-AI-BUY? •
MULTI-AGENT QUANT
ENGINE

SLIDE
02 /
10

The Solution: Adversarial Multi-Agent Council

Replacing single-prompt guesswork with institutional-grade adversarial deliberation and deterministic safety gates.

STAGE 1

Autonomous Discovery

Continuous background scanning of 40+ crypto and equity assets against quantitative liquidity filters, bid-ask spread limits, and 8-factor momentum indicators.

STAGE 2 (KEY DIFFERENTIATOR)

Adversarial Red Team

Every proposed trade is rigorously attacked by a dedicated Red Team skeptic asking: **"What if we're completely wrong?"** to expose false breakouts, wash trading, and liquidity traps.

STAGE 3

Deterministic Risk Gate

AI cannot execute trades directly. Decisions must clear a deterministic algorithmic Risk Gate enforcing max portfolio exposure, concentration caps, and position sizing limits.

● Result: Zero Emotional Errors + Fully Auditable Reasoning

Every trade requires 3+ verifiable evidence links before execution.

System Architecture & Execution Pipeline

How market data flows from Alpaca through our multi-agent council into broker execution.

01. INGESTION

Alpaca Market Data

Sub-second 1-min & 15-min crypto and equity bars, real-time quotes, and live news intelligence.

Rate Limiter: 200 req/min deduplicated

02. QUANT FUNNEL

Multi-Factor Screener

Filters candidates via liquidity (> \$100k), spread (< 100 bps), 8-factor score, and market regime.

Aging & Starvation: Rotation queue

03. AI COUNCIL

Adversarial Debate

Quant Momentum, Macro Regime, and Market Intel agents build bull case; Red Team attacks assumptions.

Featherless Open-Source LLMs

04. BROKER GATE

Risk Gate & Sizing

Computes ATR invalidation stop, sizes position, checks cash, and submits paper order to Alpaca v2.

Alpaca Paper API: Real Execution

Cloud Hosting Environment:

AWS EC2 Ubuntu 24.04 LTS (t3.micro, 2 vCPUs, 30GB gp3 SSD, 2GB Swap)

• PM2 Clustered (24/7 Daemon)

The Red Team: "What If We're Completely Wrong?"

How an adversarial agent prevents costly false breakouts and protects trading capital.

THE SKEPTIC'S AUDIT CHECKLIST

1. Liquidity Trap & Order Book Depth

Is the volume spike organic, or is the asset too illiquid for clean entry/exit without devastating slippage?

2. Macro Regime Incompatibility

Does this breakout fight against macro market trend (e.g. attempting long breakout in strong BEAR market)?

3. Asymmetric Risk/Reward Ratio

Is the expected return at least 1.25x–2.0x larger than the technical invalidation stop distance?

DELIBERATION OUTCOME MATRIX

Unlike traditional bots that binary-fire every moving average cross, our Synthesis Agent computes a consensus confidence score. If the Red Team discovers high contradictions, the candidate is instantly demoted to **HOLD** or **REJECT**.

Unanimous Strong Bull:

ENTER (100% Size Cap)

Moderate Bull + Red Team Skeptic:

SCALE (50% Sized Entry)

Unresolved Fatal Flaw / High Risk:

PASS / REJECT (\$0 Risked)

Deep Alpaca Integration: 24/7 Paper Alpha

Leveraging the full breadth of Alpaca’s modern API infrastructure for multi-asset institutional simulation.

MARKET DATA ENGINE

Multi-Asset Data Feeds

Real-time ingestion of crypto and equity historical bars, latest trades, and streaming quotes across 40+ tickers with automatic rate-limit deduplication.

EXECUTION & SIZING

Paper Trading API v2

Submits market and limit orders with deterministic client order IDs, tracking partial fills, fractional quantities, and real fill prices with zero lookahead bias.

ACCOUNT RECONCILIATION

Real-Time Portfolio Audit

Continuous sync of cash balances, buying power, cost basis, and unrealized P&L directly from Alpaca ground truth, preventing phantom fills.

ACCOUNT TESTED

\$100,000 Paper

ASSETS TRADED

Crypto & Equities

EXECUTION LATENCY

< 450ms

BROKER RECONCILIATION

Automated 15s

Production Dashboard: Complete Operator Control

No black boxes. Operators see every debate, adjust parameters on the fly, and monitor live risk metrics.

DYNAMIC RISK SLIDER

0% – 100% Capital Dial

Smooth continuous slider scaling between Standard (50% max exposure, \$5k caps) up to All-In (100% exposure, \$25k caps, 16 positions) with local persistence.

Live sync to engine via runtime API

ADVERSARIAL FEED

Real-Time Deliberation Logs

Watch the AI agents argue in real-time with full transparency: factor breakdowns, Red Team critiques, consensus voting, and reasons for rejection.

Full auditable transcript per cycle

CUSTOM TICKERS & PRESETS

Interactive Quant Ribbon

Type bar allowing users to dynamically add any crypto pair or equity ticker, saved locally with instant background Alpaca price fetching and preset manager.

Balanced / Aggressive / Conservative

Live Paper Validation: 69.2% Win Rate, 97% Deployed

Tested under real Alpaca Paper v2 execution conditions. No cherry-picked backtests, no lookahead bias.

CAPITAL DEPLOYED

96.8%

\$78.9k of \$100k active

ACTIVE POSITIONS WIN RATE

69.2%

9 out of 13 green

MAX INTRADAY DRAWDOWN

0.29%

\$294 max dip from peak

UNIT TESTS PASSED

891 / 891

100% test coverage

LIVE PORTFOLIO POSITION SNAPSHOT (BROKER AUDITED)

LTCUSD: +\$104.32 (+1.93%)

BTCUSD: +\$44.39 (+0.28%)

GOOGL: +\$19.65 (+0.74%)

FILUSD: +\$17.78 (+0.63%)

Target Market & Dual Monetization Model

Bridging the gap between \$50k/year institutional Bloomberg quant tools and 50M retail traders.

TIER 1: PRO RETAIL

\$49 / Month

Autonomous Council SaaS for active retail crypto & stock traders.

- Connect personal Alpaca API keys
- Real-time deliberation feed & alerts
- Custom risk profiles & presets

TIER 2: PROP FIRMS & QUANTS

\$299 / Month + 10%

Dedicated cloud worker instance with custom risk parameters.

- 24/7 dedicated AWS EC2 worker daemon
- Sub-second custom webhooks
- Proprietary model fine-tuning

TIER 3: B2B API

Usage-Based API

Red Team Adversarial Audit API for fintechs, wallets, and brokerages.

- "Second Opinion" API for retail brokers
- Instant adversarial risk audit per ticker
- High-margin developer API billing

Total Addressable Market (TAM):

\$14.8B Algorithmic Trading Software Market (11.2% CAGR)

Initial Focus: 12M US Retail Quant Users

The Team & Next Milestones

From hackathon MVP to commercial autonomous wealth management platform.



Nabiel

Lead Architect

Multi-Agent Council & Pipeline



Daffa

Systems & Cloud

Alpaca Broker & AWS EC2 Infra



Fortero

Quant Strategy

Factor Sizing & Risk Gates



Ghifari

Frontend & UI

Observability & Terminal UI

PHASE 1 (COMPLETED)

Alpaca Multi-Asset Engine

Multi-agent council, 24/7 AWS daemon, 891 tests passed.

PHASE 2 (Q4 2026)

Live Production Brokerage

Live money trading, options contracts, multi-broker routing.

PHASE 3 (2027)

Social Alpha & Copy Trading

Leaderboards, verified on-chain track records, mobile app.

Live Dashboard: <http://3.107.230.146:3000>

GitHub: github.com/nabielnovelkhubran/should-ai-buy